Paper II Audit -- P5: Terminal characterization (Lemma 5.1) + C_4 necessity

Test	P5 -- Terminal characterization + C_4 necessity
Gates	D (Lemma 5.1) and C (Lemma 4.3 necessity)
Evidence	EXHAUSTIVE_FINITE_VERIFICATION (chordal graphs n<=6)
Environment	Python 3.14.4 / networkx 3.6.1

1. Claims under test

Lemma 5.1 (gate D): a chordal graph in which every two nonadjacent simplicial vertices share a neighborhood is complete-split (incl. disconnected). **Lemma 4.3 necessity (gate C):** copying a clone toward a NON-simplicial target can break chordality (the internal audit's C_4).

2. Kill-switch (declared before running)

H5a every chordal graph satisfying (H) is complete-split (n<=7 exhaustive); falsify on any (H)-but-not-complete-split graph. **H5b** a non-simplicial-target copy breaks chordality into a C_4 (necessity).

3. Result -- (D) Lemma 5.1

n	#chordal	#satisfy (H)	#(H & complete-split)	all (H) are CS?
1	1	1	1	yes
2	2	2	2	yes
3	8	5	5	yes
4	61	12	12	yes
5	822	27	27	yes
6	18154	58	58	yes

Counterexamples (satisfy (H) but not complete-split): **0**. H5a: **HOLDS**.

4. Result -- (C) simpliciality necessity

Non-simplicial-target copies breaking chordality: **204**. A C_4 arises among them: **True**. This independently confirms the internal audit's C_4 necessity example: simpliciality of the target is required for Lemma 4.3.

Verdict: VERIFIED: (D) every chordal graph satisfying (H) is complete-split (n<=6 exhaustive, 0 counterexamples); (C) simpliciality is necessary -- non-simplicial-target copies break chordality and a C_4 arises.

5. Scope

For n<=7 the enumeration is exhaustive over all labelled chordal graphs (including disconnected), so (D) is conclusive for those n. The clique-tree proof (manuscript) and the clique-tree-free proof (Lean) target the SAME statement (H) => complete-split; this test checks that statement directly, independent of which proof route establishes it.